

Nexus Global HR Portal: A PWA Solution for Modern HR Management

- **Subtitle:** Module SEN5001 – Assessment PRES1
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Agenda

- The Problem
- The Solution
- Use Case Diagram
- Class Diagram
- Entity Relationship Diagram (ERD)
- solving Communication
- Solving Performance
- Backend Logic
- Database Implementation
- Conclusion & Future Work
- References

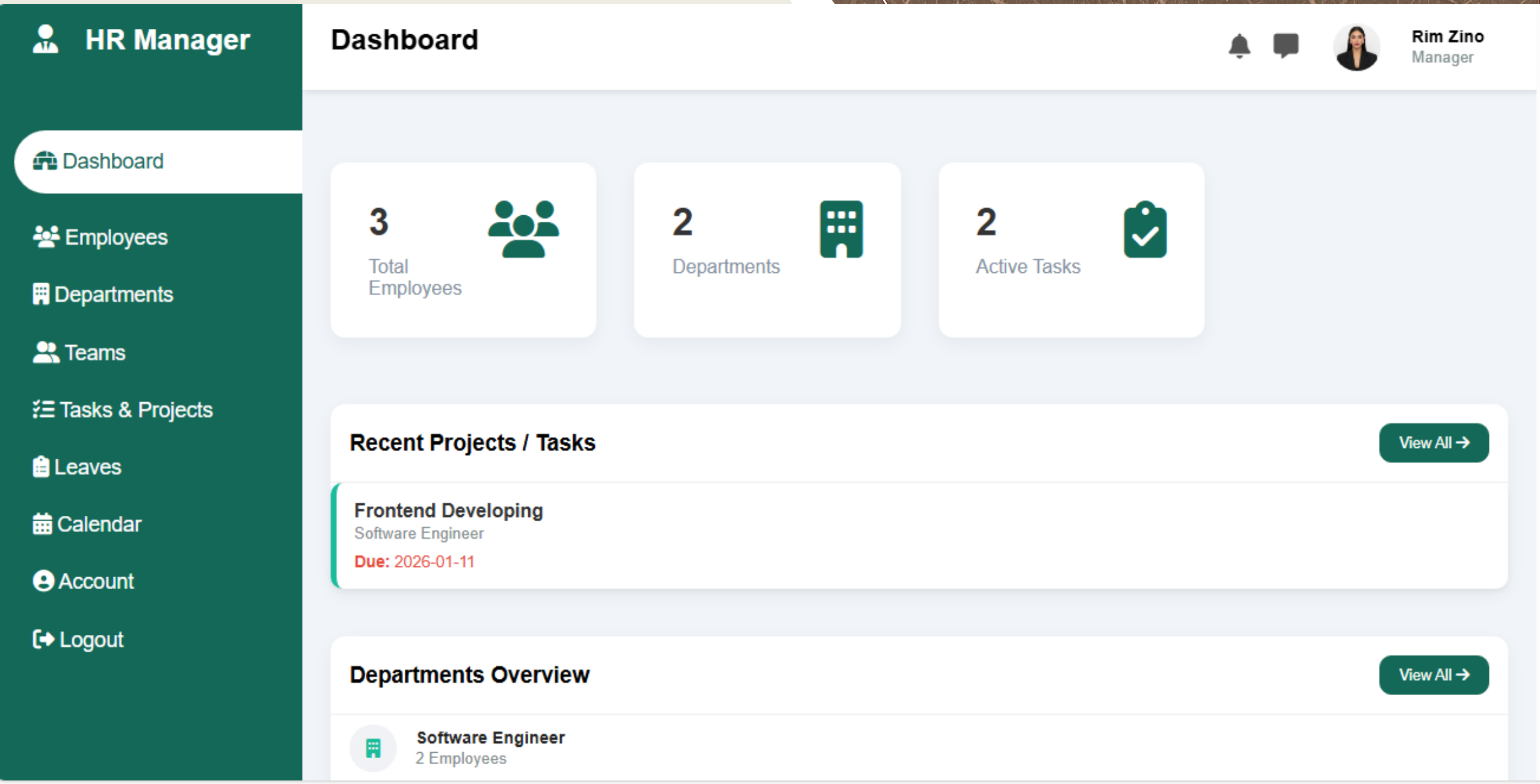


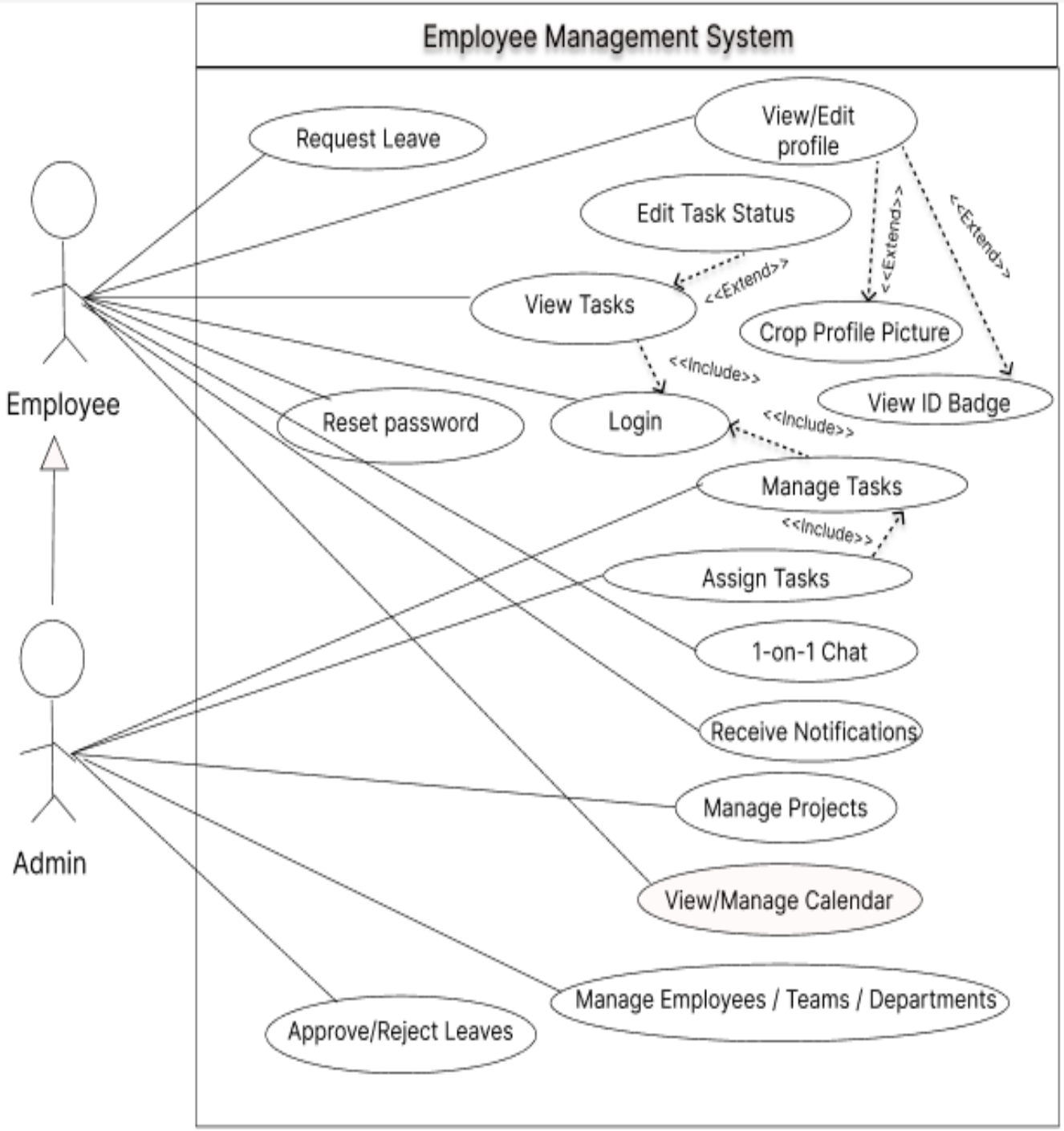
The Problem

1. Communication Gaps.
2. Manual Processes
3. Performance Tracking

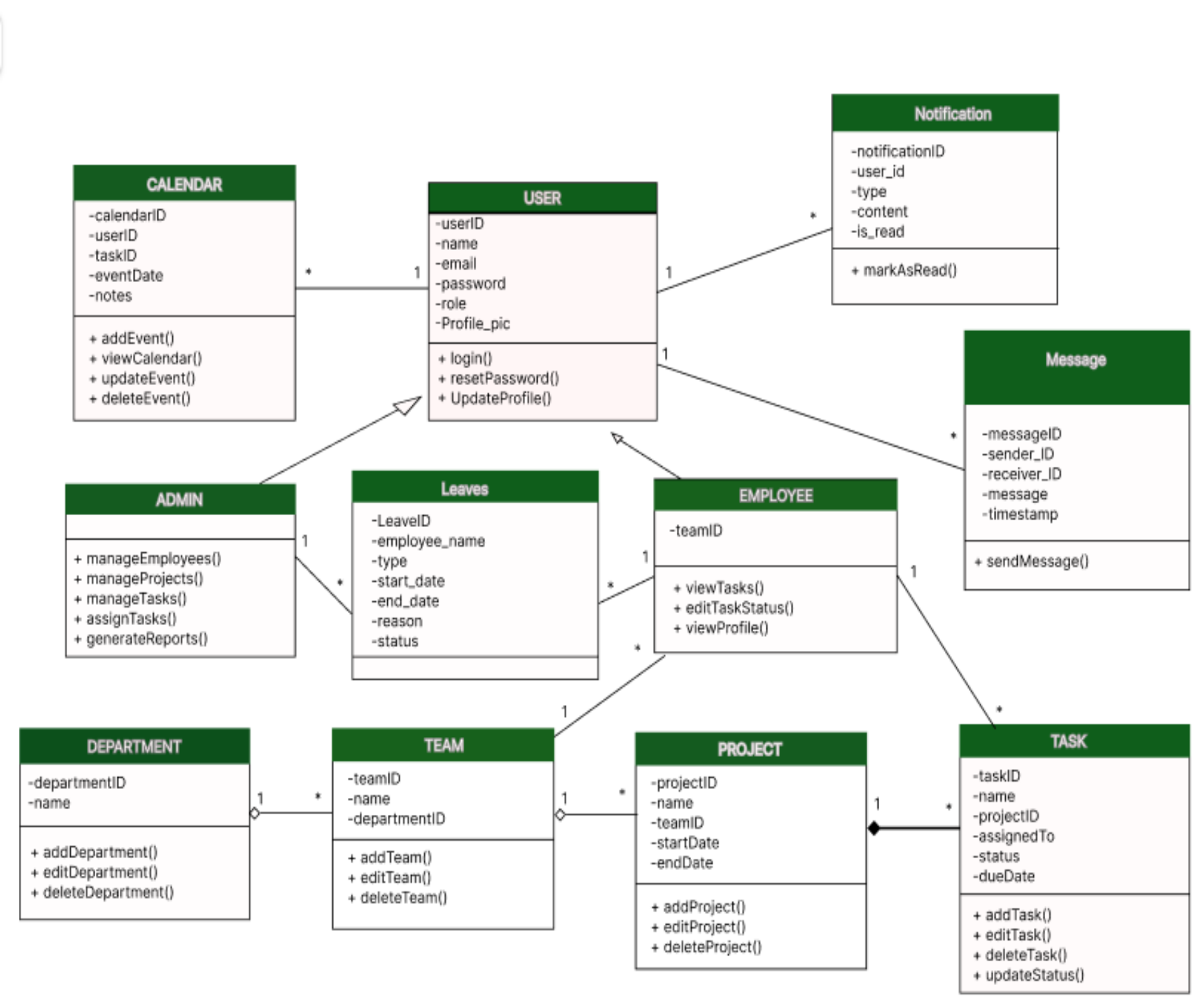


The Solution



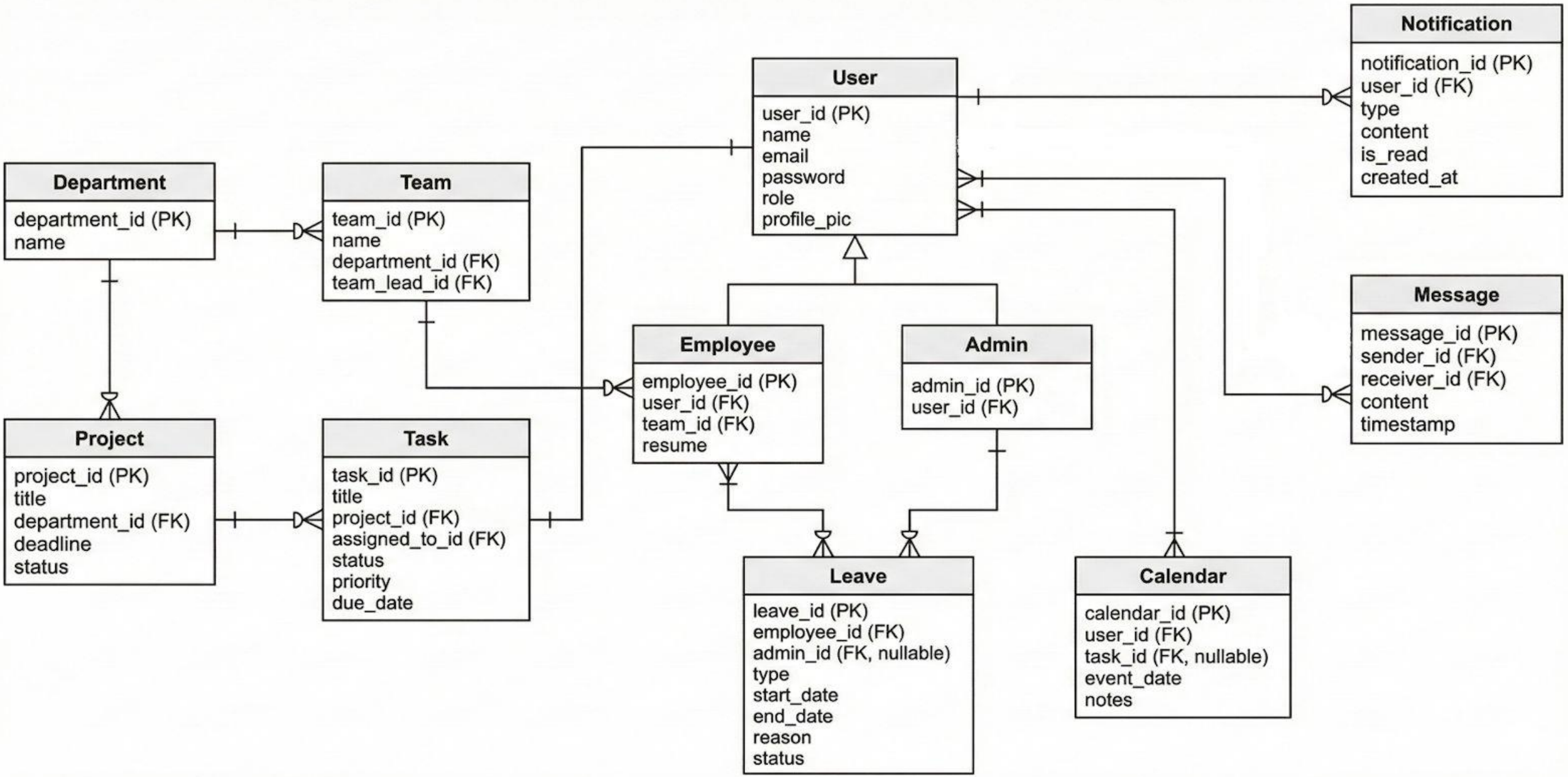


Use Case Diagram

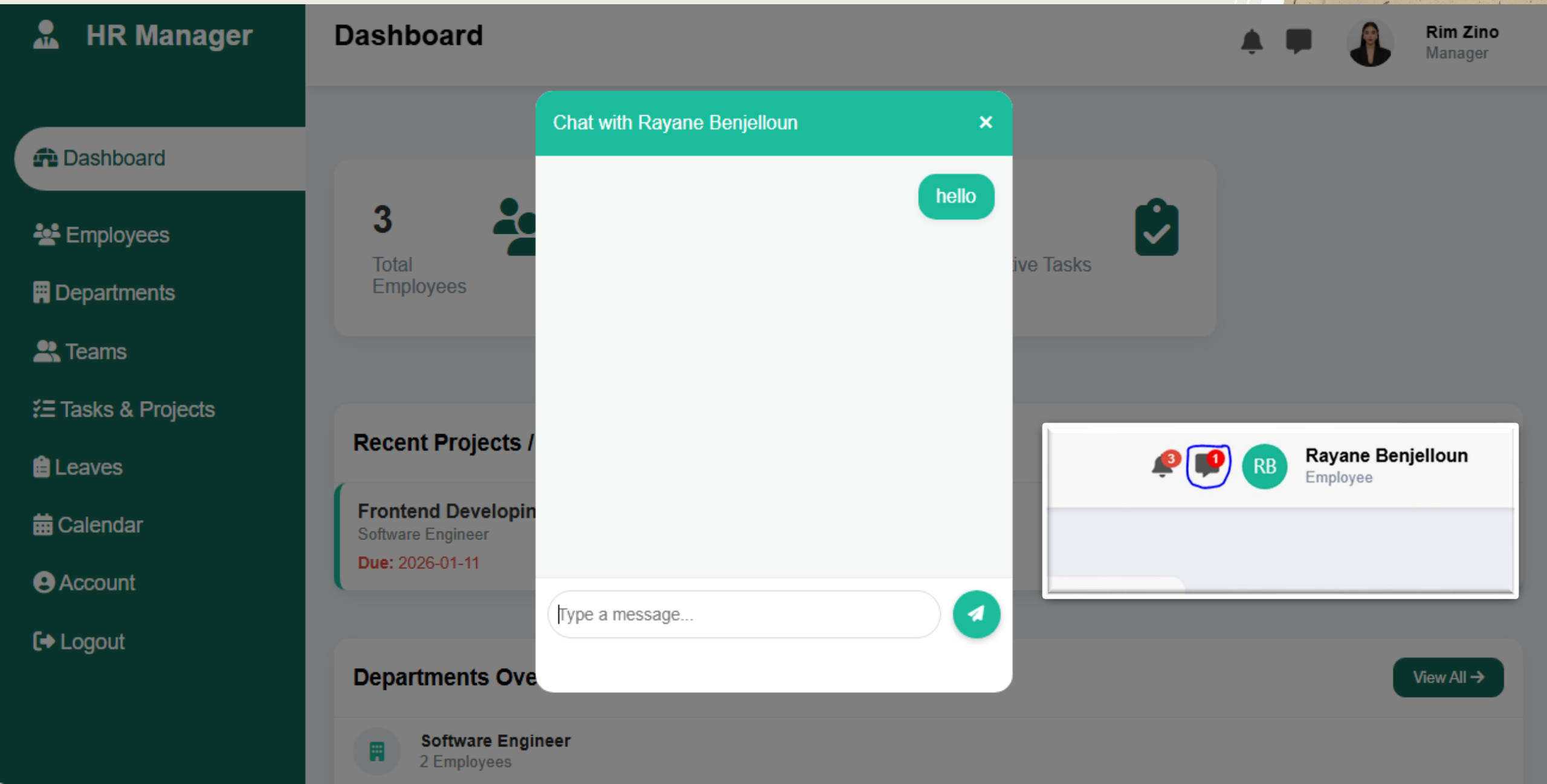


Class Diagram

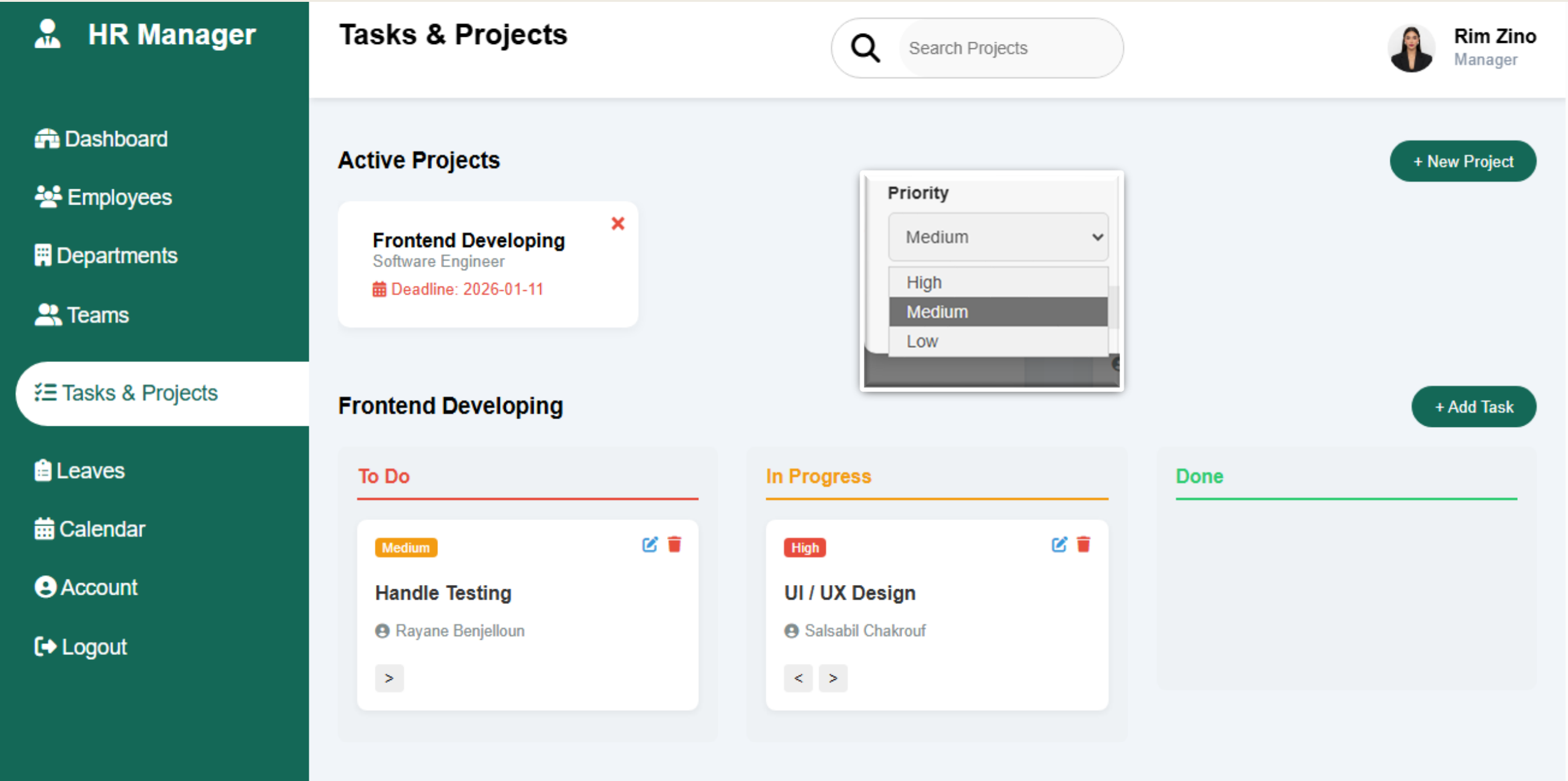
Entity Relationship Diagram (ERD)



Solving Communication



Solving Performance



Backend Logic

```
app.post('/api/login', (req, res) => {
    const { email, password } = req.body;

    db.get("SELECT * FROM users WHERE email = ?", [email], (err, user) => {
        if (err) return res.status(500).json({ success: false, message: "Database Error" });

        if (user && user.password === password) {
            res.json({ success: true, user: user });
        } else {
            res.status(401).json({ success: false, message: "Invalid email or password" });
        }
    });
});
```

```
app.post('/api/tasks', (req, res) => {
    const { project_id, title, assignee, priority } = req.body;

    db.serialize(() => {
        db.run("INSERT INTO tasks (project_id, title, assignee, priority) VALUES (?, ?, ?, ?)",
            [project_id, title, assignee, priority], function(err) {

            if (err) return res.status(500).json({ success: false });

            db.get("SELECT id FROM users WHERE name = ?", [assignee], (err, user) => {
                if (user) {
                    const msg = `You have been assigned a new task: "${title}"`;
                    db.run("INSERT INTO notifications (user_id, type, content) VALUES (?, 'Task', ?)",
                        [user.id, msg]);
                }
            });

            res.json({ success: true });
        });
    });
});
```

TABLES
- departments
- leaves
- messages
- notifications
- projects
- sqlite_sequence
- tasks
- teams
- users

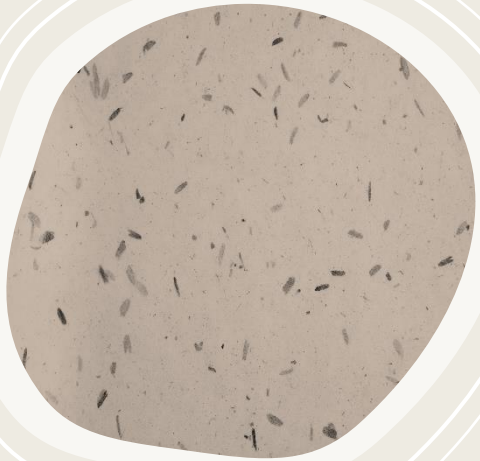
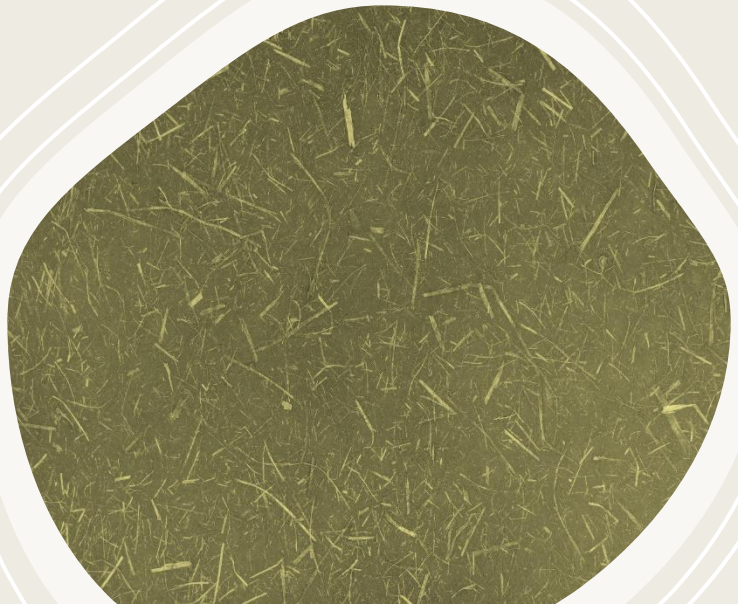
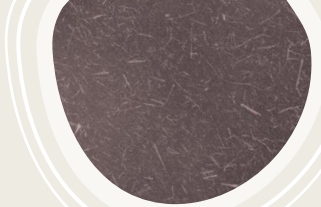
	id	name	email	password	role	departm...	status
1	1	Rim Zino	rim@gmail.com	Reem1234	Admin	Software Engine...	Active
2	2	Salsabil Chakrouf	salsabil@gmail....	salsabil	Employee	Marketing	Active
3	3	Rayane Benjello...	rayane@gmail.c...	rayane	Employee	Software Engine...	Active
4	4	Imane Sahib	imane@gmail.c...	imane	Admin	Marketing	Active
5	5	Fahd Mahfoud	fahd@gmail.com	Fahd	Employee	Software Engine...	Active
6	6	Jihane Sahib	jihane@gmail.c...	jihane	Employee	Marketing	Active
7	7	Latifa Karbal	latifa@gmail.com	latifa	Employee	Software Engine...	Active
8	8	Louay Benjelloun	louay@gmail.com	louay	Employee	Marketing	Active
9							

profile_pic	cover_pic	phone	city	zip	joined_d...	team_id	notif_prefs
NULL	NULL	NULL	NULL	NULL	2026-01-10 05:...		{"leave":tru...
NULL	NULL	NULL	NULL	NULL	2026-01-10 05:...	2	{"leave":tru...
NULL	NULL	NULL	NULL	NULL	2026-01-10 05:...	1	{"leave":tru...
NULL	NULL	NULL	NULL	NULL	2026-01-10 05:...		{"leave":tru...
NULL	NULL	NULL					
NULL	NULL	NULL					
NULL	NULL	NULL					
NULL	NULL	NULL					

Database Implementation

Rows: 6 — Filter 6 rows... — Upgrade to PRO

	id	proje...	title	assignee	status	priority
1	1	1	UI/UX	Fahd Mahfoud	In Progress	High
2	2	1	App Developement	Rayane Benjelloun	In Progress	High
3	3	1	Handle Testing	Latifa Karbal	To Do	Low
4	4	2	Competitor Analysis Report	Salsabil Chakrouf	To Do	Low
5	5	2	Design Social Media Graphics	Jihane Sahib	In Progress	Medium
6	6	2	Finalize Brand Logo	Louay Benjelloun	Done	High
7						


Conclusion & Future Work

- Conclusion & Future Work
- Future: Mobile App
- Future: Email Alerts

References

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